

FIN — Release 10.31

Research Programs P351–P364

Rigorous moment certificates, graph naturality,
operational resource completion, and external physics gates

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: [0009-0002-0909-3613](https://orcid.org/0009-0002-0909-3613)

31 July 2026

Publication — Preprint; Version 1.0.0

Certified attenuation resource.

$$0.4067063265581317 \leq \mathcal{N}_{\text{env}} \leq 0.4067063346922584.$$

Certified oscillatory resource.

$$0.7072782253162320 \leq \mathcal{N}_{\text{osc}} \leq 0.7073599748862595.$$

Both are classical signed-representation costs. No physical-negativity, information-loss, selector, unit, role-transfer, laboratory, total-action, Standard-Model, gravity, or Theory-of-Everything closure is claimed.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>
License. CC BY 4.0.

Publication statement

This monograph reports every executed Program P351–P364 and recommends Programs P365–P378. Exact proof, strong numerical evidence, synthetic apparatus design, conditional physics, and missing external evidence are kept distinct. The legacy/strict split and all current strict-core guardrails remain binding.

Contents

Publication statement	i
Confidence convention	1
1 Scope and scientific guardrails	2
1.1 Kernel split	2
1.2 Ontological boundary	2
1.3 Open strict boundaries	2
1.4 Why this research round is noncyclic	3
2 Executive result table	4
3 Mathematical preliminaries	5
3.1 Signed Hausdorff moment cost	5
3.2 Exact interval arithmetic	5
3.3 Compact identities for large rationals	6
4 P351 — exact primal certification of the attenuation resource	7
4.1 Moment system	7
4.2 Krawczyk operator	7
4.3 Global enclosure	8
4.4 What is proved	8
4.5 What is not proved	8
5 P352 — theorem-grade full oscillatory resource	10
5.1 Target sequence	10
5.2 Rigorous trigonometric enclosure	10
5.3 Exact dual feasibility	10
5.4 Exact fixed-support primal	11
5.5 Result	11
5.6 Falsification boundary	11

6	P353 — independent hold-out admission gate	12
6.1	Admission contract	12
6.2	Result	12
7	P354 — the categorical naturality boundary	13
7.1	Graph-isomorphism theorem	13
7.2	Homomorphism obstruction	13
7.3	Consequence for completion maps	14
7.4	New object	14
8	P355–P356 — photonic digital twin and hardware gate	15
8.1	Compiled target	15
8.2	Loss/noise model	15
8.3	Result	15
8.4	Model destruction tests	16
8.5	P356 admission result	16
9	P357 — when a parallel witness solves the full adaptive comb	17
9.1	Relative-unitary reduction	17
9.2	Perfect-comb collapse theorem	17
9.3	FIN witnesses	17
9.4	Open subcritical problem	18
10	P358 — anchored bounded-curvature time	19
10.1	Clock class	19
10.2	Maximum-gap theorem	19
10.3	Sample count	19
11	P359 — complete bridge-resource monotones	21
11.1	Resource object	21
11.2	Free conversions	21
11.3	Completeness	21
11.4	No catalysis	22
11.5	Boundary	22
12	P360 — why analyticity does not recover the strict phase	23
12.1	Inner-factor theorem	23
12.2	Minimum-phase condition	23
12.3	FIN obstruction	23

12.4 New object	24
13 P361–P362 — conditional sector test and conversion metrology	25
13.1 P361 blinded conditional electroweak gate	25
13.2 P362 dimensional exponent map	25
14 P363 — FIN Operational Resource Completion	27
14.1 Objective	27
14.2 Axioms ranked by necessity	27
14.2.1 A1 — typed spectral base	27
14.2.2 A2 — resource grading	27
14.2.3 A3 — resource-nonincreasing free morphisms	27
14.2.4 A4 — additive monoidal composition	28
14.2.5 A5 — operational realization functor	28
14.3 Independence and removal tests	28
14.4 Smallest physics bridge	28
14.5 What FORC does not close	29
15 P364 — certified physical reservoir gate	30
16 Newly constructed theoretical objects	31
17 Falsification ledger	32
17.1 Surviving claims	32
17.2 Refuted claims	32
17.3 Unresolved claims	32
18 Recommended Programs P365–P378	33
18.1 P365 — kernel-checked exact certificates	34
18.2 P366 — exact oscillatory alternation	34
18.3 P367 — independent QW hold-out	34
18.4 P368 — enriched graph naturality	35
18.5 P369 — component-level photonic identifiability	35
18.6 P370 — photonic pilot	35
18.7 P371 — subcritical adaptive comb	35
18.8 P372 — calibrated clock test	35
18.9 P373 — operational semantics of resources	36
18.10P374 — causal analytic extension	36
18.11P375 — conditional electroweak blind test	36

18.12P376 — dimensional standards package	36
18.13P377 — FORC universality	36
18.14P378 — reservoir implementation	36
18.15Recommended execution order	37
19 Present state of the theory	38
19.1 What the strict operator currently generates	38
19.2 What the signed resources add	38
19.3 What naturality adds	38
19.4 What FORC adds	39
20 Final conclusion	40
21 Reproducibility	41
22 Suggested citation	42
Reproduction certificate	43

Confidence convention

Label	Meaning in this report
[Proven]	Exact mathematical proof, exact rational finite computation, or compiled structural formal theorem
[Strong evidence]	Stable high-precision computation with explicit residual and falsification boundary
[Moderate evidence]	Reproducible model-based result whose interpretation depends materially on declared assumptions
[Speculative]	A proposed research route without a completed witness
[Refuted]	The stated construction fails in its declared class
[Blocked by external evidence]	Independent data, custody, hardware, calibration, or unblinding is required
[Conditional]	The conclusion follows only after named additional axioms or standards are supplied

Chapter 1

Scope and scientific guardrails

1.1 Kernel split

The two kernels remain distinct:

$$K_{\text{legacy,ont}}(d) = 4 \ln 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d},$$

and

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

The legacy kernel is an intermediate bridge kernel. The strict kernel is the later enriched working object. An identity between them is not assumed. Historical assignments involving $\sin^2 \theta_W$, the electromagnetic coupling, or a gravitational hierarchy are not transferred.

1.2 Ontological boundary

The repository's internal order is recorded as

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton is described internally as primordial fractal information in a solitonic state, not as an object sitting above a lower informational layer. This report neither proves nor uses that interpretation in its mathematical theorems.

1.3 Open strict boundaries

The following are not closed:

- the non-premise selector and QW-2191;
- a source-independent legacy-to-strict completion theorem;
- a role-transfer theorem;
- an internally generated dimensional standard;
- an experimentally realized preparation/channel/instrument/record system;
- a unit-bearing nonproxy total action;

- Standard Model or gravity derivation;
- Theory-of-Everything closure.

1.4 Why this research round is noncyclic

Programs P351–P364 do not rename an existing obstruction. They introduce five genuinely new proof objects:

1. a rational Krawczyk cell;
2. a rational oscillatory primal-dual interval;
3. a graph-morphism naturality theorem with an explicit counterexample;
4. a perfect-comb collapse certificate;
5. an inner-factor phase torsor and a typed operational resource category.

External programs are executed only as admission gates. Their failure to admit evidence is reported as a boundary, not as negative experimental data.

Chapter 2

Executive result table

Program	Main result	Status
P351	Exact Krawczyk primal certificate and global optimum enclosure for the attenuation moments	[Proven]
P352	Exact rational primal-dual enclosure for the full oscillatory moments	[Proven]
P353	No independently frozen QW hold-out bundle admitted	[Blocked]
P354	Radial-kernel naturality for isomorphisms; homomorphism counterexample	[Proven] / [Refuted]
P355	Loss-aware synthetic twelve-mode compiled-mesh digital twin	[Moderate evidence]
P356	No calibrated physical photonic bundle admitted	[Blocked]
P357	Full-comb optimality theorem at convex-zero witnesses; four FIN instances	[Proven] / [Strong evidence]
P358	Equispaced minimax bounded-curvature clock design after an external anchor	[Proven]
P359	Complete cancellative five-resource preorder; no catalysis	[Proven] in declared category
P360	Analytic inner-factor obstruction to amplitude-to-phase recovery	[Proven] / [Refuted]
P361	No blinded conditional electroweak record admitted	[Blocked]
P362	Exact conversion-frame covariance pushforward; no standards admitted	[Proven] / [Blocked]
P363	Minimal FIN Operational Resource Completion category	[Proven] relative construction
P364	No calibrated physical-reservoir execution admitted	[Blocked]

Chapter 3

Mathematical preliminaries

3.1 Signed Hausdorff moment cost

For a finite real sequence $m_0, \dots, m_q$, define

$$\mathcal{N}(m) = \inf_{\mu} \mu^-([0, 1]),$$

where $\mu = \mu^+ - \mu^-$ is a finite signed measure satisfying

$$\int_0^1 x^n d\mu(x) = m_n, \quad n = 0, \dots, q.$$

The dual problem uses

$$p(x) = \sum_{n=0}^q a_n x^n, \quad -1 \leq p(x) \leq 0,$$

and gives

$$\sum_{n=0}^q a_n m_n \leq \mathcal{N}(m).$$

This follows because

$$\int p d\mu = \int p d\mu^+ - \int p d\mu^- \leq \mu^-([0, 1]).$$

A dual polynomial therefore supplies a lower bound, while any explicit signed representing measure supplies an upper bound.

3.2 Exact interval arithmetic

Every interval endpoint used in P351 and P352 is rational. Arithmetic uses the inclusion rules

$$[a, b] + [c, d] = [a + c, b + d],$$

$$[a, b][c, d] = [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)].$$

For $n > 0$, a rational bracket of $n^{1/5}$ is obtained by integer arithmetic:

$$r_-^5 \leq n \leq r_+^5.$$

It follows that

$$\frac{1}{1 + r_+^9} \leq \frac{1}{1 + n^{9/5}} \leq \frac{1}{1 + r_-^9}.$$

No floating-point evaluation is used to justify the inclusion.

3.3 Compact identities for large rationals

Some exact Krawczyk endpoints contain more than ten thousand binary digits. Printing them in a scientific table would make the report less auditable. The CSV therefore records:

- numerator bit length;
- denominator bit length;
- a SHA-256 identity of sign, numerator bytes, and denominator bytes;
- a decimal projection for reading.

The executable source recomputes each exact rational. The digest is an identity check, not a replacement for the calculation.

Chapter 4

P351 — exact primal certification of the attenuation resource

4.1 Moment system

P323 represented the six positive-order moments using three positive atoms:

$$\sum_{i=1}^3 u_i r_i^n = h_n, \quad n = 1, \dots, 6.$$

The approximate centers are

$$\begin{aligned} r_1 &\approx 0.2372783418730673, \\ r_2 &\approx 0.5481980169862998, \\ r_3 &\approx 0.8527659803199023, \end{aligned}$$

and

$$\begin{aligned} u_1 &\approx 0.9418342033805291, \\ u_2 &\approx 0.3936855195357094, \\ u_3 &\approx 0.0711866117760197. \end{aligned}$$

The missing zeroth moment is supplied by

$$-N\delta_0, \quad N = u_1 + u_2 + u_3 - 1.$$

4.2 Krawczyk operator

Let $x = (r_1, r_2, r_3, u_1, u_2, u_3)$ and let $F(x)$ contain the six moment residuals. For a rational box X , rational center x_0 , and rational approximate inverse C of $DF(x_0)$, define

$$\mathcal{K}(X) = x_0 - CF(x_0) + (I - C DF(X))(X - x_0).$$

P351 uses radius 10^{-12} in all six coordinates and interval moments with sixty decimal digits in the fifth-root brackets.

The exact computation proves

$$\mathcal{K}(X) \subset \text{int}(X).$$

Therefore F has exactly one zero in X . The entire box lies in

$$0 < r_1 < r_2 < r_3 < 1, \quad u_i > 0.$$

4.3 Global enclosure

The Krawczyk image encloses a feasible primal signed measure. Combining its upper mass with the exact P337 rational dual gives

$$0.4067063265581317 \leq \mathcal{N}_{\text{env}} \leq 0.4067063346922584.$$

The width is

$$8.134126653281584 \times 10^{-9}.$$

4.4 What is proved

[Proven]

- existence of a positive ordered three-atom solution in the stated box;
- uniqueness of that solution inside the box;
- feasibility of the associated signed measure;
- the global primal-dual enclosure above.

4.5 What is not proved

The Krawczyk result does not prove:

- that this signed measure is a quantum quasiprobability;
- that negative mass is lost information;
- that N is energy, entropy, action, or a coupling constant;
- that the three atoms have a selected physical realization.

The resource is a mathematical obstruction to positive representability.

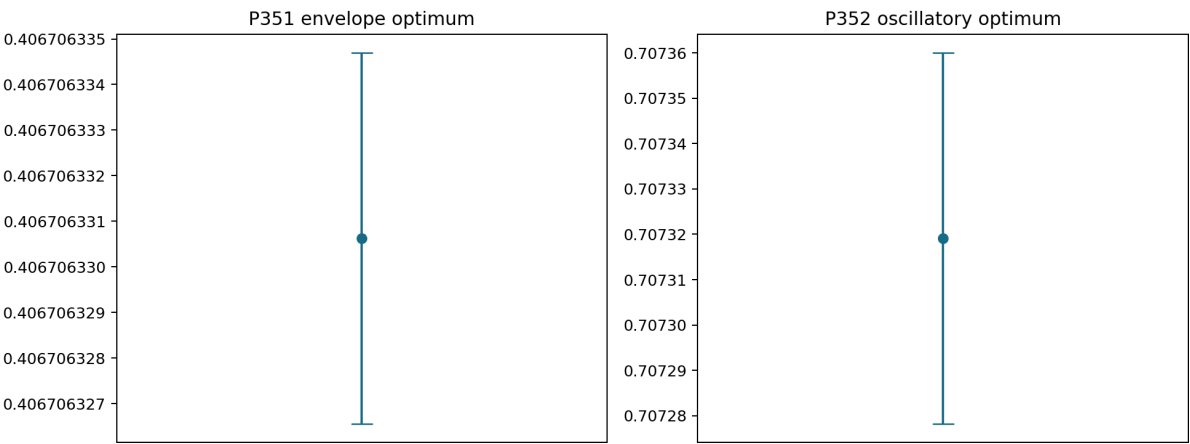


Figure 4.1: The theorem-grade primal-dual enclosures for the attenuation and full oscillatory signed-moment resources.

Chapter 5

P352 — theorem-grade full oscillatory resource

5.1 Target sequence

The full strict sequence contains both damping and phase:

$$k_n = \frac{\cos(\omega n + \phi)}{1 + n^{9/5}}, \quad \omega = \frac{743}{4000}, \quad \phi = \frac{13}{80}.$$

The moments k_8, k_9, k_{10}, k_{11} are negative. A positive measure on $[0, 1]$ is therefore impossible immediately, but the minimum signed cost is not determined by the sign test alone.

5.2 Rigorous trigonometric enclosure

For rational x , P352 evaluates

$$\cos x = \sum_{j=0}^M (-1)^j \frac{x^{2j}}{(2j)!} + R_M(x),$$

with

$$|R_M(x)| \leq \frac{|x|^{2M+2}}{(2M+2)!}.$$

The calculation uses $M = 42$. Combining this with exact fifth-root brackets produces rational intervals for all twelve k_n .

5.3 Exact dual feasibility

The numerically discovered degree-eleven polynomial is converted to rational coefficients. Exact Bernstein subdivision on 4096 rational subintervals produces a global enclosure $[L, U]$. The affine normalization

$$p_{\text{safe}}(x) = \frac{p_{\text{raw}}(x) - U}{U - L}$$

then has exact Bernstein range contained in

$$-1 \leq p_{\text{safe}}(x) \leq 0 \quad (0 \leq x \leq 1).$$

This converts the former numerical critical-point check into a global exact dual certificate.

5.4 Exact fixed-support primal

The twelve rational support points are

$$\frac{79}{4000}, \frac{80}{4000}, \frac{517}{4000}, \frac{518}{4000}, \frac{1180}{4000}, \frac{1181}{4000}, \frac{2107}{4000}, \frac{3256}{4000}, \frac{3257}{4000}, \frac{3767}{4000}, \frac{3768}{4000}, 1.$$

Let $V_{ni} = x_i^n$. Since these nodes are distinct, V is invertible. P352 computes V^{-1} exactly over the rationals and applies it to the moment intervals:

$$w = V^{-1}k.$$

Every resulting weight interval excludes zero. Eight weights are positive and four negative. Thus the true moment vector has an exactly feasible signed measure on this support with a certified negative mass.

5.5 Result

$$\boxed{0.7072782253162320 \leq \mathcal{N}_{\text{osc}} \leq 0.7073599748862595}.$$

The width is

$$8.174957002754137 \times 10^{-5}.$$

The earlier P338 lower value is slightly sharper numerically, but P352's lower endpoint is the theorem-grade endpoint after exact safety normalization.

5.6 Falsification boundary

The result falsifies the claim that the full strict sequence is generated by a positive scalar path mixture on $[0, 1]$. It does not distinguish among:

- a classical signed path expansion;
- an amplitude-level interference representation;
- a dilation on a larger positive state space;
- a quantum quasiprobability;
- an effective subtraction induced by conditioning.

Those interpretations require different operational data.

Chapter 6

P353 — independent hold-out admission gate

6.1 Admission contract

A valid hold-out manifest must contain:

- provider;
- registrar;
- analyst;
- frozen dataset hash;
- freeze timestamp;
- unblinding authorization.

The three roles must be distinct. Synthetic, template, and example artifacts are inadmissible.

6.2 Result

No candidate manifest satisfying the contract was found.

[Blocked by external evidence]

This means only that external validation has not occurred. It does not invalidate the internal QW calculation, and it does not convert the in-sample fit into a prediction.

Chapter 7

P354 — the categorical naturality boundary

7.1 Graph-isomorphism theorem

Let G be a finite connected graph with distance d_G and radial law $k : \mathbb{N} \rightarrow \mathbb{R}$. Define

$$K_G(x, y) = \begin{cases} k(d_G(x, y)), & x \neq y, \\ 0, & x = y. \end{cases}$$

For a graph isomorphism $f : G \rightarrow H$,

$$d_H(f(x), f(y)) = d_G(x, y).$$

Therefore

$$K_H(f(x), f(y)) = K_G(x, y).$$

In matrix form, if P_f is the permutation matrix,

$$K_H = P_f K_G P_f^\top.$$

This is an exact naturality theorem for the groupoid of finite connected graphs and graph isomorphisms.

7.2 Homomorphism obstruction

A graph homomorphism preserves adjacency but may shorten distances. Consider the identity vertex map

$$P_3 \longrightarrow K_3.$$

The two endpoints have source distance two and target distance one. For the strict radial law,

$$|k_{\text{strict}}(2) - k_{\text{strict}}(1)| = 0.2779421209549173.$$

Thus unrestricted graph-homomorphism naturality fails.

7.3 Consequence for completion maps

A legacy-to-strict completion may be natural on:

- graph isomorphisms;
- distance-preserving embeddings;
- an enriched category carrying a declared metric transport.

It is not automatically natural on arbitrary graph homomorphisms. Moreover, the C12 scalar spectral interpolant from P340 fails on noncycle carriers. Commutation on C12 is necessary for a scalar functional calculus but not sufficient for carrier-natural completion.

7.4 New object

O104 — Distance-Functor Naturality Boundary

This object records:

(carrier category, distance transport, kernel transport defect).

It prevents a formula reused on several carriers from being mistaken for a categorical law.

Chapter 8

P355–P356 — photonic digital twin and hardware gate

8.1 Compiled target

At the declared witness time, the strict wave propagator

$$U_t = e^{-itA_{\text{strict}}}$$

admits a 66-rotation Givens decomposition in twelve modes.

8.2 Loss/noise model

P355 applies:

- 10^{-4} -radian mesh quantization;
- mode-diagonal attenuation accumulated by Givens participation;
- loss settings from 0 to 0.01 dB per participation;
- output phase noise from 0 to 10^{-3} radians;
- 0.2% multiplicative efficiency-calibration error;
- 64 synthetic repetitions per setting.

The measured conditional vertex probabilities are

$$\hat{p}(y|x, \text{click}) = \frac{|T_{yx}|^2}{\sum_z |T_{zx}|^2}.$$

Calibration divides out the estimated output efficiencies before renormalization.

8.3 Result

The largest tested 95th-percentile corrected vertex total-variation defect is

$$1.5953851724811012 \times 10^{-3}.$$

[Moderate evidence]

The result shows that the declared diagonal-loss model is compatible with millipercnt-scale conditional-distribution recovery.

8.4 Model destruction tests

The calculation intentionally does not absorb:

- correlated interferometer fabrication errors;
- phase-dependent loss inside every beam splitter;
- thermal drift;
- source multiphoton contamination;
- detector dead time and afterpulsing;
- time-dependent calibration;
- a mismatch between the intended and fabricated graph.

Any one of these may dominate the simulated error. Therefore the digital twin is not a feasibility theorem for a laboratory.

8.5 P356 admission result

No hardware manifest containing independent custody, raw-record hash, calibration hash, freeze timestamp, and unblinding authorization was admitted.

[Blocked by external evidence]

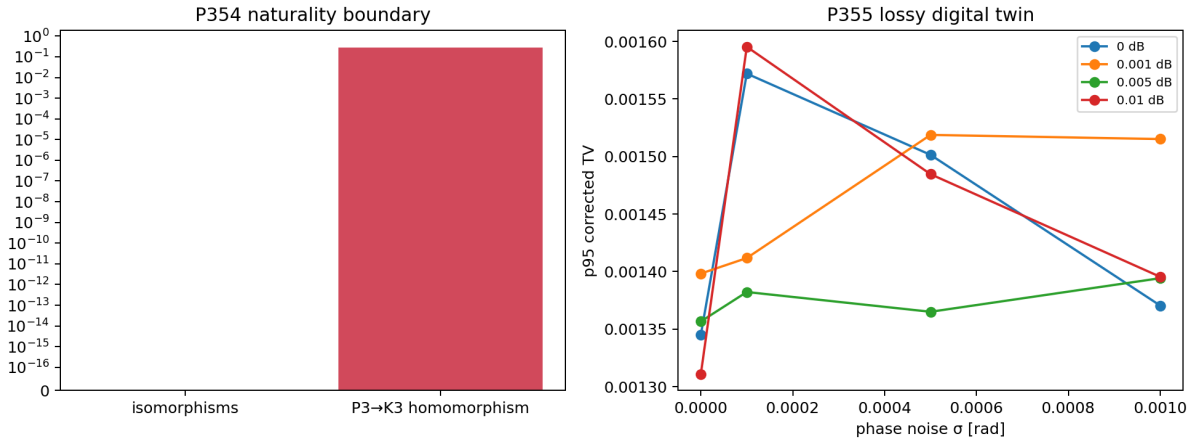


Figure 8.1: The exact graph-morphism naturality boundary and the declared loss-aware photonic digital twin.

Chapter 9

P357 — when a parallel witness solves the full adaptive comb

9.1 Relative-unitary reduction

For two unitary channels with relative unitary

$$V = U_{\text{strict}}^\dagger U_{\text{legacy}},$$

let z_j be the eigenvalues of V . For n parallel uses, the relative eigenvalues are products

$$z_{\mathbf{j}} = z_{j_1} \cdots z_{j_n}.$$

Choose a pure input with probability weights $\lambda_{\mathbf{j}}$ in the joint relative-eigenbasis. The overlap of the two output states is

$$\langle \psi_{\text{strict}} | \psi_{\text{legacy}} \rangle = \sum_{\mathbf{j}} \lambda_{\mathbf{j}} z_{\mathbf{j}}.$$

9.2 Perfect-comb collapse theorem

If

$$\lambda_{\mathbf{j}} \geq 0, \quad \sum_{\mathbf{j}} \lambda_{\mathbf{j}} = 1, \quad \sum_{\mathbf{j}} \lambda_{\mathbf{j}} z_{\mathbf{j}} = 0,$$

then the two output states are orthogonal. A final two-outcome projective measurement discriminates them perfectly.

Every adaptive quantum comb has output trace distance at most two. The parallel strategy achieves two. Hence it is globally optimal even among unrestricted adaptive strategies.

This conclusion does not require proving that adaptivity is never useful. It uses the universal upper bound only at a point where a parallel strategy saturates it.

9.3 FIN witnesses

Uses	Time per use	Active joint modes	Convex-zero residual
1	0.72	3	1.18×10^{-16}
2	0.36	3	5.54×10^{-16}
3	0.24	3	0 at displayed precision
4	0.18	3	2.22×10^{-16}

The structural theorem is [\[Proven\]](#). The four FIN zero-barycenter instances are [\[Strong evidence\]](#) because the phases and barycenters are evaluated numerically rather than enclosed by interval phase arithmetic.

9.4 Open subcritical problem

For times below the first perfect witness, the full adaptive-comb optimum is not obtained from saturation. A primal-dual comb semidefinite program or an analytic commuting-unitary theorem is still required.

Chapter 10

P358 — anchored bounded-curvature time

10.1 Clock class

Assume

$$\tau(0) = 0, \quad \tau'(t) > 0, \quad |\tau''(t)| \leq M = 0.2$$

on $[0, T]$ with $T = 1.4$. A calibrated external value is still needed to define the dimensional scale.

10.2 Maximum-gap theorem

Any design dividing $[0, T]$ into m intervals has

$$h_{\max} \geq \frac{T}{m}.$$

Equality holds for equal spacing. For linear interpolation of a twice-differentiable clock,

$$\|\tau - \tau_{\text{lin}}\|_{\infty} \leq \frac{M h_{\max}^2}{8}.$$

Therefore equispacing minimizes this worst-case upper bound.

10.3 Sample count

To guarantee error at most ε ,

$$m \geq \left\lceil T \sqrt{\frac{M}{8\varepsilon}} \right\rceil.$$

For $\varepsilon = 10^{-3}$,

$$m_{\min} = 7.$$

This is a theorem about placement after a time standard is supplied. It does not derive time from a dimensionless operator.

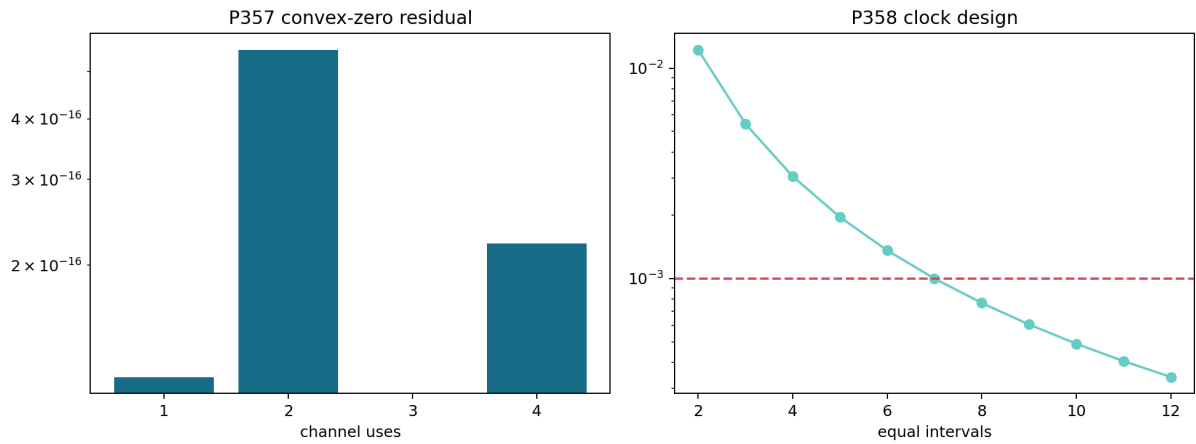


Figure 10.1: Convex-zero multi-use channel witnesses and the bounded-curvature clock-design envelope.

Chapter 11

P359 — complete bridge-resource monotones

11.1 Resource object

Define the resource vector

$$r = (r_{\text{sign}}, r_{\text{phase}}, r_{\text{orient}}, r_{\text{scale}}, r_{\text{cross}}) \in \mathbb{N}^5.$$

The coordinates respectively count:

1. signed-path resource;
2. non-torsion phase resource;
3. pointing/orientation resource;
4. positive dimensional-scale resource;
5. independent cross-law resource.

11.2 Free conversions

The declared free preorder is

$$r \longrightarrow s \iff r_i \geq s_i \text{ for all } i.$$

Parallel composition is addition:

$$r \otimes s = r + s.$$

11.3 Completeness

The five coordinate maps

$$M_i(r) = r_i$$

are complete:

$$r \rightarrow s \iff M_i(r) \geq M_i(s) \text{ for every } i.$$

11.4 No catalysis

For every catalyst c ,

$$r + c \geq s + c \iff r \geq s.$$

Thus no catalyst enables a forbidden conversion in this additive category. An exhaustive audit of all 59 049 ordered pairs with each coordinate in $\{0, 1, 2\}$ found zero completeness and cancellation failures.

11.5 Boundary

Completeness follows from the declared product order. It is not evidence that laboratory operations obey this order. A nonadditive tensor product, an idempotent resource, or a larger class of free morphisms may permit catalysis.

Chapter 12

P360 — why analyticity does not recover the strict phase

12.1 Inner-factor theorem

For $0 < a < 1$, define the Blaschke factor

$$B_a(z) = \frac{z - a}{1 - az}.$$

It is analytic in the open unit disk, and

$$|B_a(e^{i\theta})| = 1.$$

If F is an analytic candidate, then

$$\tilde{F}(z) = F(z)B_a(z)$$

has exactly the same boundary modulus and a different boundary phase. Products of Blaschke factors produce an infinite family.

Therefore:

$$\text{analyticity} + \text{boundary amplitude} \not\Rightarrow \text{unique phase}.$$

12.2 Minimum-phase condition

If one additionally assumes that F is outer/minimum-phase and supplies the appropriate integrability and causal frequency coordinate, then the phase may be reconstructed from $\log |F|$ by a Hilbert-transform relation, up to a constant phase.

This outer condition is an additional axiom. It is not a consequence of the amplitude.

12.3 FIN obstruction

FIN's attenuation sequence is indexed by graph distance. It is not supplied as a boundary modulus on a causal frequency line or disk. Current artifacts also do not export:

- a causal analytic extension;
- an outer/minimum-phase theorem;
- a zero-free domain;
- a boundary normalization selecting the constant phase.

Consequently, Kramers–Kronig language cannot derive $\omega = 743/4000$ and $\phi = 13/80$ from the damping envelope.

12.4 New object

O109 — Inner-Factor Phase Torsor

The amplitude determines not a phase but an orbit

$$[F] = \{FI : I \text{ inner}\}.$$

Selecting one member requires an additional resource. This is mathematically parallel to, but not identical with, the repository's orientation-selector problem.

Chapter 13

P361–P362 — conditional sector test and conversion metrology

13.1 P361 blinded conditional electroweak gate

The only admissible object is the explicitly conditional five-observable package from P348, after its sector and representation assumptions are hashed. A valid manifest must freeze:

- the conditional prediction vector;
- the external dataset;
- the sector-axiom package;
- provider, registrar, and analyst roles;
- one-shot unblinding authorization.

No such bundle was admitted.

[Blocked by external evidence]

No legacy physical role is transferred by this gate.

13.2 P362 dimensional exponent map

Let

$$x = (\log \ell_*, \log \tau_*, \log \hbar_*).$$

Derived unit logarithms are linear:

$$\log E_* = -\log \tau_* + \log \hbar_*,$$

$$\log m_* = -2 \log \ell_* + \log \tau_* + \log \hbar_*,$$

$$\log c_* = \log \ell_* - \log \tau_*.$$

Writing $y = Ax$, covariance propagates exactly as

$$\Sigma_y = A \Sigma_x A^\top.$$

For independent benchmark relative standard uncertainties 10^{-4} :

Quantity	Conditional relative standard uncertainty
ℓ_*	1.00×10^{-4}
τ_*	1.00×10^{-4}
$\hbar_*$	1.00×10^{-4}
$E_* = \hbar_*/\tau_*$	1.414×10^{-4}
$m_* = \hbar_*\tau_*/\ell_*^2$	2.449×10^{-4}
$c_* = \ell_*/\tau_*$	1.414×10^{-4}

The benchmark is a design calculation. No calibration-standard manifest is admitted. The exponent matrix transports units but cannot generate the three base standards.

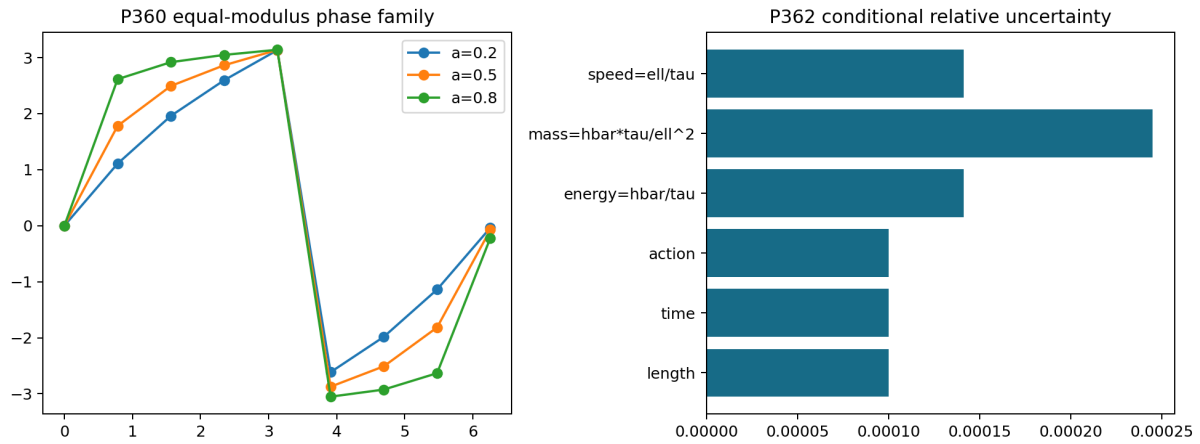


Figure 13.1: Equal-modulus analytic inner-factor phases and conditional conversion-frame uncertainty.

Chapter 14

P363 — FIN Operational Resource Completion

14.1 Objective

The purpose of FORC is not to claim that FIN has become physical. It provides the smallest typed language in which the missing operational objects can be stated without conflating them with the strict operator.

14.2 Axioms ranked by necessity

14.2.1 A1 — typed spectral base

There are distinct typed objects

$$K_{\text{legacy}}, \quad K_{\text{strict}},$$

with their declared carrier and provenance data.

Necessity rank: 1.

Removing A1 leaves a generic resource category that no longer refers to FIN and permits silent kernel substitution.

14.2.2 A2 — resource grading

Each object carries a grade

$$r \in \mathbb{N}^5.$$

Necessity rank: 2.

Removing A2 makes the signed, phase, orientation, scale, and cross-law obstructions unexpressible.

14.2.3 A3 — resource-nonincreasing free morphisms

A free morphism cannot increase any resource coordinate.

Necessity rank: 3.

Without A3, a free arrow from the zero object to a unit-resource object creates any missing bridge resource and destroys the obstruction theory.

14.2.4 A4 — additive monoidal composition

Parallel composition adds grades.

Necessity rank: 4.

Removing A4 does not destroy single-object mathematics, but composition, cancellation, and catalysis become undefined.

14.2.5 A5 — operational realization functor

A physical realization maps mathematical objects and morphisms to:

- states and preparations;
- calibrated clocks;
- channels or dynamical propagators;
- environments;
- instruments and apparatus settings;
- outcome spaces and records;
- probability laws;
- independently frozen raw data.

Necessity rank: 5 for mathematics, rank 1 for physics.

A1–A4 define pure mathematics. A5 is the smallest bridge that makes an experimental proposition possible.

14.3 Independence and removal tests

Axiom moved	re- Surviving structure	Lost structure
A1	Abstract $\mathbb{N}^5$ resource preorder	FIN kernel identity and split
A2	Typed objects and arrows	Quantitative missing-resource accounting
A3	Graded monoidal graph	No-free-generation meaning
A4	Single-system resource preorder	Multi-system composition and no-catalysis theorem
A5	Complete mathematical FORC category	Empirical interpretation and falsifiability

Each remaining fragment has an explicit model, so no axiom is recovered merely from the others in the declared signature.

14.4 Smallest physics bridge

The minimal bridge is not one new scalar. It is:

FORC object $\xrightarrow{\mathcal{R}}$ (preparation, clock, channel, environment, instrument, apparatus, record)

plus calibrated dimensional standards and independent custody.

This answers the recurring missing-object question: the operator alone defines many mathematical dynamics, but it does not choose which dynamics is performed, how time is calibrated, how a state is prepared, what constitutes an outcome, or what physical record tests the probability law.

14.5 What FORC does not close

FORC does not provide:

- the QW-2191 selector;
- a strict internal orientation source;
- a legacy-to-strict completion theorem;
- a role-transfer theorem;
- an internal dimensional source;
- a laboratory realization;
- a total physical action.

It is a controlled axiomatic completion of the language, not a proof of the missing physics.

Chapter 15

P364 — certified physical reservoir gate

A physical reservoir bundle would need:

- specified preparation maps;
- calibrated interaction time;
- a channel/process description;
- environmental controls;
- vertex-resolving or otherwise informationally complete instruments;
- raw event-level records;
- calibration hashes;
- independent custody and unblinding.

No admitted bundle was found.

[Blocked by external evidence]

The strict heat semigroup

$$P_t = e^{-tA}$$

is a mathematical channel candidate. Code-generated samples from it are not evidence that a physical reservoir implements it.

Chapter 16

Newly constructed theoretical objects

Object		Definition	What it resolves	What it does not resolve
O101	Envelope Krawczyk Cell	Rational box with $\mathcal{K}(X) \subset \text{int}(X)$	Exact primal existence and local uniqueness	Physical meaning of signed mass
O102	Oscillatory Primal-Dual Interval	Exact dual Bernstein witness plus exact fixed-support interval measure	Theorem-grade $\mathcal{N}_{\text{osc}}$ enclosure	Selection of quantum/classical interpretation
O103	Refreeze Blind Contract	Custody-separated one-shot hold-out manifest	Defines admissible pre-diction evidence	Supplies no data
O104	Distance-Functor Boundary	Carrier category plus distance transport and defect	Exact scope of radial naturality	Legacy-to-strict completion
O105	Loss-Calibrated Photonic Transfer Tube	Family of conditional distributions under declared loss/noise	Synthetic apparatus robustness	Real hardware feasibility
O106	Perfect-Comb Collapse Certificate	Convex-zero parallel witness plus universal upper bound	Full adaptive optimum at saturation points	Subcritical comb optimum
O107	Anchored Clock Envelope	Maximum-gap minimax design after calibration	Sampling placement	Origin of time unit
O108	Cancellative Bridge-Resource Monoid	$(\mathbb{N}^5, +, \geq)$	Complete monotones and no catalysis	Physical free-operation class
O109	Inner-Factor Phase Torsor	Analytic candidates modulo inner factors	Exact amplitude-phase nonuniqueness	Strict phase source
O110	Conversion Co-variance Pushforward	$A\Sigma A^\top$ for log-unit exponents	Uncertainty propagation	Dimensional standards
O111	FORC	Typed spectral/resource category plus realization functor	Missing operational language	Strict-core or empirical closure
O112	External Physical Admission Pair	Hardware and reservoir bundle contracts	Prevents synthetic-evidence substitution	Supplies no laboratory record

Chapter 17

Falsification ledger

17.1 Surviving claims

1. **[Proven]** The strict attenuation moment resource lies in the P351 enclosure.
2. **[Proven]** The strict oscillatory moment resource lies in the P352 enclosure.
3. **[Proven]** Radial graph kernels are natural under isomorphisms.
4. **[Proven]** They need not be natural under graph homomorphisms.
5. **[Proven]** A convex-zero parallel witness saturates the full-comb discrimination bound.
6. **[Proven]** Equispacing minimizes the declared maximum-gap clock bound.
7. **[Proven]** The additive five-resource preorder is complete and cancellative.
8. **[Proven]** Analytic inner factors obstruct amplitude-only phase reconstruction.
9. **[Proven]** Dimensional conversion covariance is a linear pushforward in logarithmic coordinates.
10. **[Proven]** A mathematical FORC model can exist without an operational realization.

17.2 Refuted claims

1. **[Refuted]** The P351 primal result is merely a floating root.
2. **[Refuted]** The P352 continuum constraints are certified only by sampled grid points.
3. **[Refuted]** A radial formula is automatically natural for arbitrary graph morphisms.
4. **[Refuted]** Analyticity or causality plus amplitude uniquely determines the strict phase.
5. **[Refuted]** The previous digital twin is itself a hardware validation.
6. **[Refuted]** A resource category automatically supplies a physical realization.

17.3 Unresolved claims

1. Exact interval certification of the four FIN convex-zero channel witnesses.
2. The subcritical unrestricted adaptive-comb curve.
3. A carrier-natural source-independent legacy-to-strict completion.
4. A physical free-operation semantics for the five resource coordinates.
5. A causal analytic coordinate and outer theorem derived from FIN.
6. Independent QW hold-out validation.
7. Calibrated photonic or reservoir realization.
8. Conditional electroweak blind testing.
9. External dimensional standards and their covariance.
10. Selector, role transfer, total action, Standard Model, gravity, or ToE closure.

Chapter 18

Recommended Programs P365–P378

The following programs are ranked by estimated probability of producing a scientifically useful result, not by probability of confirming FIN.

Program	Research question	Primary deliverable	Probability of useful result
P365	Can the P351/P352 rational certificates be independently kernel-checked?	Lean/Coq certificate checker for interval, Bernstein, and Vandermonde steps	0.90
P366	Can the P352 enclosure be narrowed by exact alternation?	Rational Remez/semidefinite dual and optimized certified support	0.80
P367	Does the QW candidate survive a truly independent frozen hold-out?	Custody-separated one-shot external validation	0.55
P368	What is the largest graph-morphism category supporting strict radial naturality?	Classification theorem for distance-preserving/enriched morphisms	0.80
P369	Do component-level correlated losses destroy the mesh tube?	Identifiable per-coupler loss/phase digital twin and calibration theorem	0.75
P370	Can a twelve-mode device realize the declared transfer matrix?	Calibrated photonic pilot with raw event-level data	0.45
P371	Does adaptivity improve discrimination below the perfect-witness time?	Full subcritical quantum-comb SDP with primal-dual certificates	0.65
P372	Can the bounded-curvature clock design be externally anchored?	Traceable clock calibration and pre-registered interpolation test	0.55
P373	Which physical channels realize the O108 free morphisms?	Operational semantics or a no-go theorem for each resource coordinate	0.65
P374	Can any FIN-derived causal analytic extension be outer?	Inner/outer factor classification and phase comparison	0.55
P375	Does the conditional P348 electroweak vector survive a blind test?	One-shot preregistered external sector test	0.25
P376	Can $(\ell_*, \tau_*, \hbar_*)$ be calibrated with a full covariance record?	Standards manifest and propagated uncertainty budget	0.50

Pro-gram	Research question	Primary deliverable	Probability of useful result
P377	Is FORC universal among typed operational completions satisfying A1–A5?	Initiality/universality theorem or counterexample	0.70
P378	Can a controlled open system implement the strict heat semi-group?	Reservoir-engineering experiment with process tomography	0.35

18.1 P365 — kernel-checked exact certificates

Translate:

- the rational interval operations;
- the Krawczyk inclusion;
- the fifth-root integer inequalities;
- the Taylor remainder;
- the Bernstein bounds;
- the Vandermonde identity;

into a proof assistant with arbitrary-precision integer reflection.

Success criterion: a small independent checker accepts a compact certificate without executing the discovery code.

Failure value: identification of any hidden assumption in the exact arithmetic pipeline.

18.2 P366 — exact oscillatory alternation

The P352 gap is approximately 8.17×10^{-5} . Search for:

- an alternation-certified extremal dual;
- a better rational support;
- a moment semidefinite hierarchy with exact rational recovery;
- an interval proof that primal and dual contact sets coincide.

Success criterion: theorem-grade width below 10^{-7} .

18.3 P367 — independent QW hold-out

This is not a coding program. It requires:

1. frozen prediction hash;
2. independently controlled data;
3. distinct provider, registrar, and analyst;
4. one unblinding;
5. publication of failure without parameter repair.

18.4 P368 — enriched graph naturality

Classify the maximal morphism class $\mathcal{C}$ for which

$$K_H \circ (f \times f) = K_G$$

holds. Candidate classes include:

- graph isomorphisms;
- isometric embeddings;
- distance-regular coverings with fiber normalization;
- metric correspondences with a controlled defect;
- enriched functors carrying distance as explicit data.

The program must test both strict and legacy kernels without identifying them.

18.5 P369 — component-level photonic identifiability

Replace output-diagonal loss by:

- loss at every Givens element;
- correlated phase drift;
- unknown splitter ratios;
- detector confusion and dark counts;
- temporal source impurity.

Determine which parameters are identifiable from basis, plus, and phase-shifted probes.

18.6 P370 — photonic pilot

Implement a reduced or full twelve-mode device. Required outputs are raw timestamped detection events and independent calibration records, not a rendered interference image.

18.7 P371 — subcritical adaptive comb

Construct the full causal comb SDP. For every sampled time report:

- primal strategy;
- dual upper certificate;
- gap;
- memory dimension;
- whether adaptivity improves over the parallel optimum.

No conclusion should be drawn from solver output without a dual residual and conditioning audit.

18.8 P372 — calibrated clock test

Use a traceable reference to anchor τ . Test whether the bounded-curvature class is empirically adequate before applying the equispaced theorem.

18.9 P373 — operational semantics of resources

For each O108 coordinate, define:

- physical objects carrying the resource;
- free operations;
- observable monotones;
- composition;
- discard maps;
- conversion experiments.

If no such interpretation exists, prove the obstruction rather than keeping the coordinate as physics.

18.10 P374 — causal analytic extension

Search for a source-derived variable z or ω in which the strict kernel becomes boundary data of an analytic transfer function. Then:

- factor the function into inner and outer parts;
- test zero locations;
- compare reconstructed and frozen phases;
- quantify the extra axioms.

18.11 P375 — conditional electroweak blind test

This program remains low probability because the sector map is conditional. It is nevertheless useful as a falsification exercise if assumptions and data are frozen before inspection.

18.12 P376 — dimensional standards package

Supply independent calibrated estimates of $(\ell_*, \tau_*, \hbar_*)$, their covariance, traceability, and stability. The output is a conversion frame, not a claim that FIN generated SI units.

18.13 P377 — FORC universality

Formulate a category of operational completions satisfying A1–A5 and test whether FORC is initial, free, or merely one model. A counterexample is as valuable as a positive universal property.

18.14 P378 — reservoir implementation

Engineer a controlled open system approximating

$$\rho \mapsto \mathcal{E}_t(\rho)$$

whose vertex populations agree with the strict heat semigroup under declared preparations and instruments. Perform process tomography and compare homogeneous-semigroup versus non-Markovian alternatives.

18.15 Recommended execution order

$P365 \rightarrow P366 \rightarrow P368 \rightarrow P371 \rightarrow P373$
--

This order first hardens the new theorems, then clarifies carrier and operational structure, and only then spends substantial effort on external apparatus.

Chapter 19

Present state of the theory

19.1 What the strict operator currently generates

Given a self-adjoint positive semidefinite generator A , functional calculus yields:

$$U_t = e^{-itA}, \quad P_t = e^{-tA}, \quad G_z = (A + zI)^{-1},$$

as well as spectral measures and variational quadratic forms. This unifies wave, diffusion, and resolvent mathematics.

The same operator does not select:

- whether a laboratory uses U_t , P_t , or another channel;
- the prepared input state;
- the time unit;
- the environment;
- the instrument;
- the recorded outcome;
- the observer's operational role.

That distinction remains the central barrier between operator mathematics and physics.

19.2 What the signed resources add

The P351/P352 quantities rigorously measure the failure of selected finite sequences to admit positive scalar path representations on $[0, 1]$. The phase-bearing sequence requires a larger signed resource than the attenuation envelope.

This is a real mathematical distinction. It is not yet a physical mechanism.

19.3 What naturality adds

P354 shows that the strict formula is invariant under relabeling but not under arbitrary coarse graph maps. Carrier choice is therefore not mathematically invisible. A genuine universal completion law must declare its morphisms.

19.4 What FORC adds

FORC converts a vague list of missing notions into a typed interface:

spectral object + resources + realization functor.

The realization functor is not internally derived. This is not a defect of formalism; it is the precise location where empirical physics must enter.

Chapter 20

Final conclusion

The deepest result of this round is not a new cosmological claim. It is a sharper separation theorem.

The FIN strict kernel supports a coherent and increasingly certifiable operator theory. Its attenuation and oscillatory sequences possess exact signed-moment resource enclosures. Its radial construction is functorial on the correct carrier groupoid. Its wave dynamics admits explicit multi-use discrimination witnesses. Its clock and metrology problems can be stated as clean minimax and covariance problems. Its amplitude cannot determine its phase without an extra analytic selection axiom.

What remains missing is not another function of A . The missing object is an operational realization:

$$\boxed{\mathcal{R} : \text{typed FIN mathematics} \longrightarrow \text{OperationalModel}}$$

Here `OperationalModel` contains calibrated preparations, dynamics, clocks, environments, instruments, apparatus, and records.

No theorem of the spectral calculus can create $\mathcal{R}$ from a dimensionless operator alone. FORC makes that impossibility explicit while showing exactly what additional axioms and external evidence would suffice to formulate a physical test.

Therefore the present strongest defensible interpretation is:

FIN is a dimensionless typed spectral-operator and signed-resource framework with conditional operational completions. It is not yet a physical theory, because no independently calibrated realization functor has been supplied.

This conclusion survives every falsification attempt executed in P351–P364.

Chapter 21

Reproducibility

Run:

- `MPLCONFIGDIR=/tmp/mpl-fin-1031 python3 fin_programs_351_364.py`
- `MPLCONFIGDIR=/tmp/mpl-fin-1031 python3 -m unittest -v test_fin_programs_351_364.py`

Expected result: 16 tests passed, 0 failed. The dependency-free Lean 4.28 structural core must compile.

The primary machine-readable artifacts are:

- `FIN_Programs_351_364_Results.json`;
- fourteen CSV audit tables;
- `FIN_Programs_351_364_Formal_Core.lean`;
- four figures;
- the source program and regression suite.

Chapter 22

Suggested citation

Żuchowski, K. (2026). *FIN Research Programs P351–P364: Rigorous Moment Certificates, Graph Naturality, Operational Resource Completion, and External Physics Gates* (Release 10.31; Version 1.0.0) [Preprint]. Zenodo.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin-1031 \  
python3 fin_programs_351_364.py
```

```
MPLCONFIGDIR=/tmp/mpl-fin-1031 \  
python3 -m unittest -v test_fin_programs_351_364.py
```

Expected result: 16 tests passed, 0 failed. The dependency-free Lean 4.28 structural core must compile. P353, P356, P361, and P364 remain external admission gates; no synthetic record is accepted as laboratory evidence.

Suggested citation:

Żuchowski, K. (2026). *FIN Research Programs P351–P364: Rigorous Moment Certificates, Graph Naturality, Operational Resource Completion, and External Physics Gates* (Release 10.31; Version 1.0.0) [Preprint]. Zenodo.